Experience Study AI Copilot

Route actuarial data prep, A/E analysis, and visualization requests through the orchestrator.

Profile the raw synthetic inforce data and list the data types for MAC, MEC, MAF, and MEF.

I profiled the raw file data/input/synthetic_inforce.csv. No data transformations were applied and no output file was written.

Summary of the profiling run

- Source file: data/input/synthetic_inforce.csv
- Total rows: 15,008
- Unique Policy_Number count: 5,000
- Columns discovered: Policy_Number, Gender, Issue_Age, Smoker, Face_Amount, Risk_Class, Duration, MEC, MAC, MOC, MEF, MAF, COLA
- Memory use: ~5.01 MB

Data-type and null status (requested fields)

- MAC: dtype = float64, nulls = 0
- MEC: dtype = float64, nulls = 0
- MAF: dtype = float64, nulls = 0
- MEF: dtype = float64, nulls = 0

Notable data-quality observation

- COLA has 14,971 nulls (expected in life insurance when no COLA benefit exists — this is not an error per the domain rule).

Next steps I can run (tell me which you want)

- Run actuarial data checks (MAC/MEC/MOC logic, exposure checks)
- Create age or face-amount bands and save the analysis dataset to data/output/analysis_inforce.csv
- Regroup categorical risk classes

If you want a saved analysis dataset or specific checks, tell me which operation and any banding parameters; I will perform them and save the output to data/output/analysis_inforce.csv.

Group Face_Amount into 4 equal-width bands, and then group Issue_Age into 4 equal-width bands.

Done — I created 4 equal-width bands for Face_Amount and 4 equal-width bands for Issue_Age, saved to data/output/analysis_inforce.csv.

What I did

- Read data from data/input/synthetic_inforce.csv.
- Created a new categorical column Face_Amount_band with 4 equal-width bins spanning the min–max of Face_Amount.
- Created a new categorical column Issue_Age_band with 4 equal-width bins spanning the min–max of Issue_Age.
- Saved the transformed dataset to data/output/analysis_inforce.csv (did not drop or impute any rows).

Notes and assumptions

- Binning used equal-width (range divided into 4 intervals). Bin edges are computed from the min and max of each source column.
- Nulls in Face_Amount or Issue_Age were preserved; rows with Null in a source column have a Null band value.
- Per your domain guardrail, I did not alter COLA or any other expected Nulls.

If you want

- The exact bin edges and counts per band, I can provide a summary.
- Different banding (quantiles or custom edges) or labeling conventions (e.g., inclusive/exclusive edges, left/right closed) — say which and I'll re-run.

Run a 1-way dimensional sweep on Gender, Risk Class, and Face_Amount_band to calculate the A/E ratio by Count and Amount.

1-way dimensional sweep complete on the prepared analysis dataset using Gender, Risk_Class, Face_Amount_band, ranked by AE_Ratio_Amount.

Top 10 ranked cohorts

Rank	Cohort	AE_Ratio_Amount	AE_Ratio_Count	Sum_MAC
1	Risk_Class=Standard Plus	2.06	1.76	13
2	Face_Amount_band=0.0	2.04	1.60	13
3	Face_Amount_band=2.0	1.88	1.58	9
4	Risk_Class=Standard	1.80	1.38	14
5	Gender=M	1.60	1.53	26
6	Face_Amount_band=1.0	1.15	1.04	8
7	Face_Amount_band=3.0	1.13	0.91	7
8	Gender=F	1.13	0.90	11
9	Risk_Class=Preferred Plus	0.99	1.00	5
10	Risk_Class=Preferred	0.38	0.75	5

- Worst cohort: `Risk_Class=Standard Plus`
- Financial risk (AE_Ratio_Amount): 2.06
- Selection/mortality risk (AE_Ratio_Count): 1.76
- Count 95% CI: [0.99, 2.91]
- Amount 95% CI: [1.15, 3.41]
- Actuarial interpretation: Signal is directionally adverse, but statistical certainty is limited (wide CI and/or interval crossing 1.0).
- Cohorts evaluated in output: 10

Run 2-way dimensional sweeps for all pairs between Smoker, Risk_Class, Gender, and Issue_Age_band, then rank the results by AE_Ratio_Amount.

2-way dimensional sweep complete on the prepared analysis dataset using Smoker, Risk_Class, Gender, Issue_Age_band, ranked by AE_Ratio_Amount.

Top 10 ranked cohorts

Rank	Cohort	AE_Ratio_Amount	AE_Ratio_Count	Sum_MAC
1	Smoker=Yes | Risk_Class=Preferred Plus	13.08	8.06	2
2	Smoker=Yes | Issue_Age_band=1	9.44	6.99	2
3	Smoker=Yes | Gender=M	3.21	2.20	2
4	Risk_Class=Standard Plus | Issue_Age_band=3	2.94	2.43	7
5	Risk_Class=Preferred Plus | Issue_Age_band=1	2.83	1.89	2
6	Risk_Class=Standard | Issue_Age_band=0	2.82	1.93	2
7	Risk_Class=Standard | Issue_Age_band=2	2.80	2.38	6
8	Risk_Class=Standard Plus | Issue_Age_band=2	2.54	1.90	4
9	Gender=M | Issue_Age_band=2	2.46	2.21	10
10	Risk_Class=Standard | Gender=M	2.27	1.70	10

- Worst cohort: `Smoker=Yes | Risk_Class=Preferred Plus`
- Financial risk (AE_Ratio_Amount): 13.08
- Selection/mortality risk (AE_Ratio_Count): 8.06
- Count 95% CI: [1.69, 24.98]
- Amount 95% CI: [2.74, 40.54]

- Actuarial interpretation: Signal is directionally adverse, but statistical certainty is limited (wide CI and/or interval crossing 1.0).
- Cohorts evaluated in output: 20

Generate a treemap of the 2-way sweep on Gender and Smoker

Treemap report generated and opened: /Users/amberwang/Desktop/dzw/experience-study-ai-copilot/data/output/temp_treemap_report.html

Run 2-way dimensional sweeps for all pairs between Smoker, Gender then rank the results by AE_Ratio_Amount.

2-way dimensional sweep complete on the prepared analysis dataset using Smoker, Gender, ranked by AE_Ratio_Amount.

Top 10 ranked cohorts

Rank	Cohort	AE_Ratio_Amount	AE_Ratio_Count	Sum_MAC
1	Smoker=Yes | Gender=M	3.21	2.20	2
2	Smoker=No | Gender=M	1.50	1.49	24
3	Smoker=No | Gender=F	1.20	0.95	11

- Worst cohort: `Smoker=Yes | Gender=M`
- Financial risk (AE_Ratio_Amount): 3.21
- Selection/mortality risk (AE_Ratio_Count): 2.20
- Count 95% CI: [0.46, 6.91]
- Amount 95% CI: [0.67, 10.13]
- Actuarial interpretation: Signal is directionally adverse, but statistical certainty is limited (wide CI and/or interval crossing 1.0).
- Cohorts evaluated in output: 3

Generate a treemap of the 2-way sweep we just ran

Treemap report generated and opened: /Users/amberwang/Desktop/dzw/experience-study-ai-copilot/data/output/temp_treemap_report.html

Generate a treemap for Smoker and Risk Class

Unable to generate treemap: the current sweep artifact does not contain the requested pair `smoker + risk_class`. Available pairs in `data/output/sweep_summary_2_smoker_gender.csv`: gender + smoker.

Ask the copilot to profile data, run a sweep, or generate a chart.